

Offline-First PWA for Teaching Informatics in Wartime Conditions: Architecture and Survey Results

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Abstract — The paper focuses on the design and evaluation of an offline-first Progressive Web App (PWA) for teaching informatics in wartime conditions, where online learning takes place amid recurring disruptions to Internet connectivity and power supply. The proposed solution is based on a layered architecture: a client-side PWA with local progress persistence and an event queue; an offline/edge layer built on service workers and caching; and an online layer that supports synchronization, AI-based hints, and teacher analytics. Pedagogical support is implemented as a stepwise scaffolding model that strengthens learners' autonomy while limiting the delivery of ready-made solutions; the motivational layer emphasizes resilience through non-competitive mechanisms of individual progress. The empirical evaluation was conducted through surveys of pupils ($n = 33$), teachers ($n = 10$), and experts ($n = 5$), using a five-point Likert scale and open-ended questions. The analysis combined descriptive statistics (M , SD) with a thematic synthesis of comments. The results show high ratings for usability under unstable connectivity and for alignment with the needs of war-affected border regions. Pupils rated the multi-level hints and AI support particularly highly, whereas teachers provided more differentiated assessments of this component. Experts confirmed the strengths of the offline-first approach and identified priorities for further improvement, including more intuitive navigation and expanded teacher reporting tools (notably, exporting results).

Keywords — *offline-first, PWA, AI tutor, low-connectivity learning, resilient education, informatics education, gamification, teacher preparation*

I. INTRODUCTION

The full-scale war has turned digital support for school education from a “convenient add-on” into a precondition for maintaining continuity of teaching and learning in many communities. In several regions, online instruction amid unstable internet access and rolling power outages has become routine, directly undermining the feasibility of synchronous lessons and the reliability of traditional web services. UNICEF, for example, documents situations in which children are forced to learn online amid recurrent connectivity and power outages [1]. At the same time, air-raid alerts and the need to interrupt lessons due to security threats systematically erode instructional time, with public reports recording substantial losses of class hours caused by air-raid sirens over the academic year [2]. Under these conditions, a digital tool for teaching informatics cannot

follow the same logic as platforms designed for permanent connectivity. If a web application depends on a stable connection, every network interruption disrupts the learner's activity, causes loss of intermediate results, and reduces motivation to continue. In contrast, an offline-first approach treats the network as an optional resource: the application remains functional without internet access and synchronizes when connectivity becomes available. This is why Progressive Web App (PWA) architectures are promising for low-connectivity scenarios: they combine the reach of a web platform with offline capabilities through a service worker, caching, and controlled request handling [3].

Technically, the core of offline-first behavior in a PWA is the service worker, which acts as a programmable proxy between the application and the network, intercepting requests, returning cached responses, and providing a “baseline experience” without an internet connection [3]. In practice, this usually requires a carefully designed caching strategy (for both static assets and data) to reduce dependence on network conditions and keep essential features available offline [4]. At the same time, the PWA architecture must align with educational use cases. Different parts of the application have different value in offline mode, so it is important to identify a minimal set of features that must always work (viewing materials, completing tasks, storing progress) and separate them from those that can be deferred until synchronization (reporting, content updates, analytics, etc.) [5].

This paper presents an offline-first PWA for teaching informatics under wartime conditions and reports the findings of a survey involving three stakeholder groups (students, teachers, and experts) who evaluated how well the solution fits the constraints of real educational settings in Ukraine. The study has two aims: (1) to describe the architectural principles and key technical mechanisms that keep the PWA platform operational in the absence of internet connectivity; and (2) to empirically examine how different stakeholders perceive the usability, reliability, and usefulness of the PWA platform for organizing informatics instruction under unstable connectivity and frequent lesson interruptions.

The contribution of this work lies in combining two components: (a) a description of an offline-first PWA architecture tailored to educational scenarios under systemic disruptions, with a clear separation between

offline and online functionality; and (b) the presentation of stakeholder evaluation results, which make it possible to compare the expectations and needs of students, teachers, and experts and to identify directions for improving the PWA platform based on their feedback.

II. RELATED WORK

UNICEF's analytical materials on digital learning emphasize that access is determined not only by device availability but also by real barriers to meaningful connectivity, including data costs, unstable connections, and constraints in home infrastructure [6]. In its guidance on digital learning in emergencies, UNICEF specifically highlights the need for solutions that operate under low-connectivity conditions and support sustainable models of delivering and updating learning materials without constant internet access [7]. The UNICEF Innocenti "Off to learn" report further discusses practical approaches to organizing offline content updates and reducing data usage in low-access environments [8].

In parallel, international policy has increasingly focused on public digital platforms as an infrastructure that can be adapted to different access conditions. UNESCO's initiative on public digital learning platforms ("Gateways") frames the platform as a public good rather than a collection of isolated services [9]. In addition, the UNESCO-UNICEF Charter formulates principles for the development of such platforms, stressing equitable access, safety, and quality [10]. For our purposes, this is important for two reasons: first, the requirement for access must explicitly include offline scenarios; second, technical solutions must align with pedagogical and ethical frameworks when a platform processes children's data and supports instructional decision-making.

The technical line of research and development that most directly supports learning under unstable connectivity is associated with offline-first web application architectures, particularly the PWA approach. The Service Workers standard defines a model of an "event mediator" between the network and the application, allowing developers to intercept requests, manage caches, and implement offline behavior [3]. Practical guidance from MDN clarifies that a common "cached assets first" strategy can provide a basic offline experience and, when possible, refresh data from the network [11]. Web.dev training materials similarly emphasize that Cache Storage and well-designed caching strategies reduce reliance on network conditions and make an application suitable for intermittent connectivity, which aligns with the needs of educational scenarios in crisis-affected regions [4]. In this sense, PWAs offer a practical technological framework for implementing the principle that "learning activities do not stop when the network goes down," provided that the architecture clearly separates critical offline functions (viewing materials, completing tasks, saving progress locally) from deferred online operations (content updates, analytics transmission, results synchronization).

A related strand of work concerns intelligent support for learners. Research on the effectiveness of intelligent tutoring systems shows that computer-based tutors can produce substantial learning gains, although effect sizes

depend on system design and implementation conditions [12]. The landscape is now complicated by the emergence of generative models that can sustain dialogue and provide hints, but also introduce risks of incorrect answers, overreliance, and data leakage. As a result, ethical and governance frameworks have moved to the forefront. UNESCO's "Guidance for generative AI in education and research" formulates principles intended to curb uncontrolled use of generative AI, including requirements for data protection, transparency, and pedagogical appropriateness [13]. These principles are consistent with UNESCO's broader standard-setting on AI ethics, which underscores the primacy of human rights, dignity, fairness, and human oversight in AI applications [14]. At the policy level, OECD likewise analyzes how countries are developing safeguards around generative AI in education (privacy, intellectual property, risks of misuse), reinforcing the case for a cautious design of AI components in educational platforms [15].

In informatics education, it is particularly important to distinguish between a hint that scaffolds thinking and a hint that replaces it. Recent work on cognitive offloading with digital technologies stresses that delegating part of cognitive processing to external tools can improve immediate performance, but requires careful instructional design to avoid reducing the depth of learning [16]. Taken together, these related studies lead to two practical conclusions for our work: an offline-first PWA is a technically well-grounded response to unstable connectivity, and AI-based support should be tightly governed and bounded so as to preserve the educational value of tasks and avoid creating additional risks.

III. DESIGN REQUIREMENTS AND PEDAGOGICAL RATIONALE

Designing a PWA platform for teaching informatics in wartime conditions must proceed from the assumption that it has to operate under low-connectivity conditions, support offline execution of learning activities, and correctly restore state after interruptions [7]. Accordingly, the baseline requirement is formulated as offline-first: core learning scenarios should function without internet access, and result synchronization should be deferred until a connection becomes available. In a PWA, this behavior is technically grounded in the service worker, an event-driven component that can intercept requests and return cached resources, keeping the application usable offline [3]. From this follow two concrete sub-requirements.

First, a caching strategy is needed for both static assets (interface, core modules, help) and learning materials that must remain available without a network connection. Web.dev guidance emphasizes that Cache Storage reduces reliance on network conditions and provides the basis for offline availability, provided that caching is designed around realistic usage scenarios [4]. Second, the platform must manage offline data (progress, answers, activity logs). In practice, this typically involves combining a cache for assets with a separate storage mechanism for structured data, since an offline experience must not only "open a page without the network," but also preserve learner outputs with reliable recovery after the application is restarted [17].

The second group of requirements concerns integrity and fairness in recording progress. Under unstable connectivity, the platform cannot treat the moment of sending data to the server as the moment when a learning action is performed. Instead, it is reasonable to log events locally (time, action type, task identifier, response state), accumulate them in a queue, and synchronize later. This model reduces the risk of losing results due to a dropped connection and improves activity tracking in situations where the learner works in short bursts between air-raid alerts or power outages.

The third group of requirements relates to data usage and performance. In low-connectivity environments, the platform should favor a text-first presentation, limit heavy media, and rely on progressive loading and small incremental content updates. Practical PWA guidelines emphasize the need to distinguish between what is “critical for initial use” and what “can be loaded later,” as this distinction largely determines how reliable the application feels under weak network conditions [4].

The fourth group of requirements is pedagogical, and concerns support for the learner’s thinking rather than substituting for their problem-solving. This is particularly important in informatics education: hints should encourage error analysis, clarification of the problem statement, and breaking a task into steps. If an AI component is used, it should operate in a controlled scaffolding mode, with constraints on providing complete solutions or ready-made code. Here, it is appropriate to align with normative principles for public digital platforms, where access and quality must be combined with safety and accountability, especially when the platform processes children’s educational data [10].

The fifth group of requirements concerns platform motivation. Under heightened stress, competitive and comparative mechanisms (for example, leaderboards) may amplify pressure and disengagement, whereas individual progress indicators, activity streaks, and “small achievements” can support regular engagement without additional stress. This is not merely a “gamification option,” but part of an engineering-pedagogical strategy for sustaining learning activity when external conditions systematically interfere with schooling.

In the following sections, we describe how these requirements were implemented in the architecture of the offline-first PWA and present survey findings on users’ experience with the platform.

IV. SYSTEM ARCHITECTURE AND IMPLEMENTATION

The PWA platform was implemented as a web application designed to remain functional without network access and to resume operation quickly after interruptions. Technically, the solution is built on Next.js (App Router) with a separation between server and client components. This separation allows moving only the minimal necessary computation to the server, while keeping interactivity, the code editor, and local persistence on the client side. This aligns with the Server/Client Components model in Next.js and supports a flexible combination of static rendering, caching, and dynamic interface fragments [18]. As shown

in Fig. 1, the platform implements the offline-first approach through a three-layer organization. In the client layer, the learner’s activity is concentrated: completing tasks, working with the code editor, and locally storing states and events. The offline/edge layer (service worker and cache) ensures that the core content can be launched and accessed without a network and provides an offline fallback, while the online layer handles result synchronization, AI-based hints, and generation of learning analytics for the teacher. This separation allows core learning scenarios to remain functional during connectivity interruptions while preventing data loss through deferred synchronization once the connection is restored.

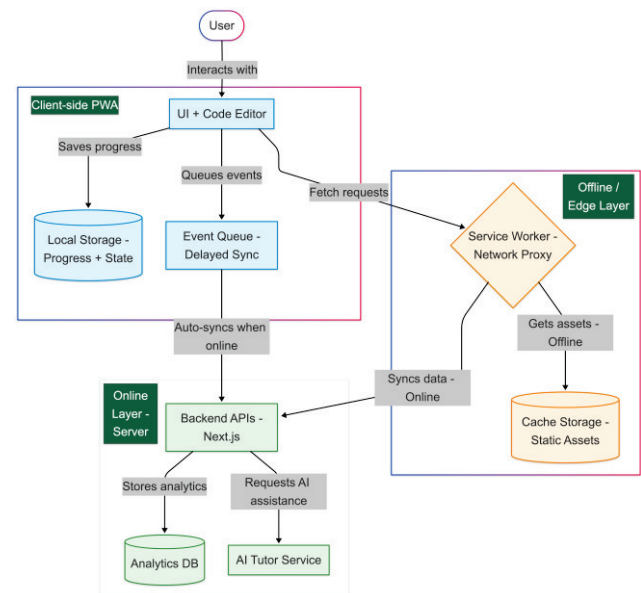


Figure 1. Architecture of the offline-first PWA platform

The service worker plays a central role in enabling autonomy. It intercepts network requests and returns cached responses, allowing the application to start and continue functioning even when the internet is unavailable. The service worker specification describes this model as an event-driven intermediary between the application and the network that supports caching and request management [3]. For educational use cases, the guiding principle is that, in offline mode, actions must support learning activity (opening a lesson, reading materials, completing a task, saving progress), whereas content updates and transmission of aggregated data can be postponed (caching is treated not as an “optional extra” but as the default reliability mechanism [19]). At the interface level, an explicit online/offline status indicator is provided so that learners understand whether their work is being stored locally or has already been synchronized.

Because connections often drop without warning during wartime, the system records learning actions locally. This is implemented via the Web Storage API (local storage), which persists data across browser sessions. As a result, code entered by the learner, intermediate answers, and task completion states are not lost after a page reload or device restart [20].

Synchronization is implemented in a deferred manner: local events (e.g., task completion, an attempt, a received

hint) are appended to a queue and sent to the server when the network becomes available. Crucially, the moment the action is performed is decoupled from the moment the data are delivered. This design means that a learner's progress is not "penalized" by technical failures and that activity logs are more accurate in situations where learning is repeatedly interrupted by air-raid alerts or power cuts.

Communication with the AI tutor is via streaming responses: the learner sees the answer appear incrementally rather than waiting for the full message to be generated. In low-bandwidth environments, this is critical, as it reduces the likelihood that a short outage will nullify the hint's benefit. Vercel's documentation recommends streaming as a way to serve model responses, and the AI SDK illustrates how the interface can start rendering output before the response is fully computed [21]. Importantly, the AI module is an optional layer rather than a precondition for platform functionality: when the network is unavailable, the learner continues working with materials and tasks, and hints simply become available again once connectivity returns.

The engineering choices are further oriented toward saving bandwidth and energy: the interface favors "lightweight" representations, and critical resources are loaded first. In Next.js, this is supported by a controlled combination of server-side rendering and client-side interactivity, which helps keep initial load times short even on low-performance devices [18].

The AI component is designed as a controlled support mechanism for problem-solving rather than a generator of complete solutions. Its hint logic aligns with research on intelligent tutoring systems, which shows that learning gains depend strongly on the quality of help-seeking and the structure of assistance, rather than on the mere availability of on-demand hints [22].

The support model is implemented as a sequence of help levels, where each subsequent level increases specificity without replacing the learner's own thinking: (1) clarifying questions (checking understanding of the problem statement and input/output data); (2) a hint about the underlying principle or method (which algorithmic structure or language construct is appropriate); (3) stepwise guidance (outlining a solution plan and identifying sub-tasks); (4) error localization (explaining why the current approach produces an incorrect result and what needs to be changed). This "hint ladder" supports self-regulation: the learner can stop at an early level if that is sufficient to move forward independently, without having to progress to more detailed hints unless necessary [22].

Because generative AI can produce inaccurate or pedagogically inappropriate responses, the hint module is embedded with safeguards: (a) prompts are formulated to prioritize explanations and questions rather than complete solutions; (b) "end-to-end" solutions in the form of full code for typical tasks are disallowed; (c) the use of hints is logged as an event (number of requests, levels of help), which enables more accurate interpretation of learning outcomes and supports integrity policies. Functionally, the AI tutor operates only when a network connection is available and does not affect core offline functionality: when the internet is unavailable, the learner continues

working with materials and tasks with local progress storage, and hints become accessible again once connectivity is restored.

The platform's motivational design is conceived as a means of supporting regular engagement and returning to learning after forced interruptions, rather than as a mechanism for competition. This aligns with theoretical work showing that the effects of gamification depend on which elements are used and which psychological needs they reinforce (competence, autonomy, relatedness) [23]. The platform relies primarily on non-competitive approaches: indicators of personal progress (progress bars, completed topics/steps) as feedback on competence; achievements/badges for effort and persistence (e.g., completing a series of exercises, returning after a break, maintaining clean code), which reinforce a sense of advancement; activity streaks and "soft recovery" after gaps (without penalty logic), to reduce demotivation when interruptions are caused by external factors.

Leaderboards and rankings are not central to the learning experience, as research in educational settings shows they can have mixed effects: for some learners, they increase engagement, but for those at the bottom of the rankings, they may reduce interest and motivation [24]. Instead, priority is given to mechanics that support individual learning trajectories and avoid creating a stable hierarchy of "winners and those who fall behind." The choice of elements is driven not by "game for the sake of the game," but by a pedagogical function: to support self-regulation, a sense of progress, and willingness to continue learning under unstable conditions [25].

V. RESEARCH DESIGN

The evaluation of the platform was organized as a pilot study to assess perceptions across different stakeholder groups. Data were collected during pilot implementation in 2025 in the frontline city of Shostka (Sumy region, Ukraine). The survey involved 33 school learners (grades 7–11, mean age 14.7 years, 53% male), 10 teachers (more than 5 years of school teaching experience, including at least 3 years of distance teaching), and 5 experts (2 informatics teachers, 2 university instructors in IT-related disciplines with at least 10 years of experience, and 1 IT professional/software developer). Learners and teachers evaluated the system from the standpoint of everyday classroom use, while experts assessed the platform from an engineering-pedagogical perspective, including its suitability for unstable infrastructure and its potential for scaling.

For learners and teachers, data were collected using questionnaires with a 5-point Likert scale (from 1 – "definitely not" to 5 – "definitely yes") and one open-ended question on how the platform could be improved. After completing typical actions on the PWA platform (viewing materials, working on tasks, receiving hints, and, when internet access was available, interacting with the AI tutor), respondents filled out the questionnaires. Data collection was anonymous; only ratings and textual comments were recorded. Responses were analyzed at the item level: for each statement, the mean (M) and standard deviation (SD) were calculated. To aid interpretation of practical

significance, the proportion of scores 4–5 was additionally used as an indicator of positive perception. Open-ended responses were analyzed qualitatively using thematic coding, followed by grouping recurrent suggestions for improvement (interface and navigation, offline scenarios, hint/AI policies, teacher tools, reporting).

This pilot study focused on perceived usability, acceptability, and contextual fit of the PWA platform rather than on comparing learning outcomes.

VI. RESULTS

Descriptive statistics for the three stakeholder groups are presented below (Table I). Learners rated support through hints/AI and comfort of working in unstable conditions the highest (for both items, 100% of responses were 4–5). Ratings of the interface and gamification remained positive but showed greater variability ($SD \approx 1.03$ – 1.09), indicating a heterogeneous user experience and differences in how motivational mechanics are perceived. In open-ended comments, learners repeatedly pointed to the value of the offline mode (“it’s convenient that it works without internet”) and to problems attributed specifically to connectivity (“only the internet was lagging”), which aligns with the offline-first logic: the critical bottleneck is not task completion but the availability of the online extensions of the PWA platform.

TABLE I. STAKEHOLDER SURVEY RESULTS

Group	Indicator	M±SD	4–5, %
Learners	Clarity and accessibility of the interface	3.94±1.09	73
	Usefulness of multi-level hints and the AI assistant	4.70±0.47	100
	Motivational effect of gamification	4.00±1.03	73
	Comfort of learning under unstable internet/stress	4.73±0.45	100
Teachers	Tools for uploading and managing materials	4.10±0.57	90
	Monitoring presence and progress	4.50±0.53	100
	Usefulness of AI tools for differentiated support	4.10±0.88	70
	Fit with frontline-region conditions	4.70±0.48	100
Experts	Functionality and technical implementation	4.20±0.33	85
	Adaptability to frontline conditions	4.80±0.18	100
	Usefulness for the educational process	4.20±0.30	93

Teachers rated the platform’s fit with frontline-region conditions and the progress-tracking module highly (both items received 100% of scores in the 4–5 range). Evaluations of AI tools showed the greatest variability ($SD = 0.88$) and a lower share of 4–5 scores (70%), which is typical for an innovative component: teachers differ in how they define appropriate use cases and boundaries for hints. Two clear requests for future versions emerged from the open-ended responses: exporting grades/results to a single Excel file to reduce manual work with the gradebook, and a dedicated module for dictation tasks with markers/indicators showing whether a text/answer was generated independently by the learner or with the help of generative AI.

Experts gave the highest possible ratings to offline operation with synchronization ($M = 5.0$), psychological adequacy of the design ($M = 5.0$), and the uniqueness of the

overall approach ($M = 5.0$). The lowest means were recorded for intuitiveness of the interface/navigation ($M = 3.8$), stability under unstable internet ($M = 4.0$), and adequacy of analytical reports ($M = 4.0$). This points to a clear improvement trajectory: UX/navigation, the quality and granularity of reporting, and the robustness of online extensions under weak connectivity. In open-ended comments, experts most frequently emphasized the value of offline learning and the combination of the PWA approach with hints, while also noting the risk of overreliance on hints/AI. As a recommendation, they suggested scaling to humanities subjects, provided that the offline-first logic and the controlled nature of support are preserved in the PWA platform.

VII. DISCUSSION

The findings are consistent with the PWA platform’s design logic as an offline-first solution: learners and teachers gave their highest ratings to the ability to work and learn under unstable connectivity (100% of ratings at 4–5 for the corresponding items). This confirms the validity of separating core offline scenarios from online extensions (Fig. 1): when learning activities do not depend on the network, interruptions cause fewer breakdowns in activity and less cognitive effort is spent on resuming work, which aligns with the technical properties of PWAs and existing guidance on digital learning in crisis settings [3, 4, 11]. At the same time, the gap between high ratings for “resilience” and more variable ratings of the interface and gamification among learners is telling. It suggests that a reliable offline core is a necessary but insufficient condition for a uniformly positive user experience: for some learners, navigation and the “visibility” of support elements determine whether the platform feels simple and motivating. Expert ratings, where interface intuitiveness received the lowest scores, reinforce the need to prioritize UX refinement.

Results on AI-based hints illustrate the different perspectives of learners and teachers. Learners rate multi-level support as highly useful, which is consistent with research on intelligent tutoring systems showing that well-structured help can effectively support task completion [12]. Teachers’ more cautious attitudes reflect concerns about autonomy, transparency, and academic integrity. Against this background, the chosen scaffolding model – stepwise levels of help, a ban on full ready-made solutions, and logging of help requests – appears both pedagogically appropriate and aligned with international recommendations on responsible use of generative AI in education [13, 14].

The motivational layer, designed to support persistence rather than competition, received generally positive – though not unanimous – ratings from learners, while experts assessed the overall gamification approach favorably. This aligns with studies showing that gamification effects depend on how specific elements are embedded into learning activities and which psychological needs they support [25, 26]. In a high-stress context with frequent interruptions, priority is given to tracking individual progress and “soft return” mechanisms rather than leaderboards and stable rankings, which can create additional pressure, especially for learners who consistently

appear at the bottom of the list [24]. The requests from teachers and experts regarding reporting point to a practical direction for further development. Teachers emphasize the need to export results to spreadsheets and reduce manual work in gradebooks, while experts call for richer analytical reports. In crisis conditions, such tools are not optional additions but components of the platform's organizational resilience alongside its technical robustness.

The study has clear limitations: it is based on small, non-random samples and focuses on perceptions and user experience rather than on learning gains compared to alternative tools. The results should therefore be interpreted as indicative rather than representative of all war-affected schools, yet they provide concrete guidance for improving the PWA platform.

VIII. CONCLUSIONS

This paper presents the architecture of an offline-oriented PWA for teaching informatics during wartime and reports the results of a pilot stakeholder survey conducted in a frontline city in Ukraine. The platform is built around an offline-first principle, ensuring that core learning scenarios remain available without network access, while synchronization and online extensions are activated once connectivity is restored. Survey results indicate high acceptability of the platform under unstable conditions.

Future work will focus on expanding the set of teacher-facing tools (including export of results to spreadsheet formats and richer reporting with information about the types of hints used), improving UX navigation and the visibility of key functions for different user groups, and empirically confirming the platform's impact on learning outcomes and learner autonomy through quasi-experimental studies. An additional direction is scaling the approach to other school subjects, where offline accessibility and deferred synchronization are equally critical.

REFERENCES

- [1] 'UNICEF distributes learning kits for distance learning in Ukraine | UNICEF'. Accessed: Dec. 30, 2025. [Online]. Available: <https://www.unicef.org/ukraine/en/stories/unicef-distributes-learning-kits-for-distance-learning>
- [2] 'Ukraine: Air raid sirens halted one in every five lessons this school year - Ukraine | ReliefWeb'. Accessed: Dec. 30, 2025. [Online]. Available: <https://reliefweb.int/report/ukraine/ukraine-air-raid-sirens-halted-one-every-five-lessons-school-year>
- [3] 'Service Workers Nightly'. Accessed: Dec. 30, 2025. [Online]. Available: <https://www.w3.org/TR/service-workers>
- [4] 'Caching', web.dev. Accessed: Dec. 30, 2025. [Online]. Available: <https://web.dev/learn/pwa/caching>
- [5] 'Architecture', web.dev. Accessed: Dec. 30, 2025. [Online]. Available: <https://web.dev/learn/pwa/architecture>
- [6] M. Brossard et al., 'Digital learning for every child: closing the gaps for an inclusive and prosperous future'. [Online]. Available: <https://www.unicef.org/media/113896/file/Digital%20Learning%20for%20Every%20Child.pdf>
- [7] 'UNICEF Digital Learning in Emergencies: Frameworks, Approaches, Initiatives', [Online]. Available: <https://surl.it/rpnvdd>
- [8] 'Off to learn: Making offline digital learning work for vulnerable girls in Mauritania'. [Online]. Available: <https://www.unicef.org/innocenti/media/3491/file/UNICEF-Off-to-Learn-EN.pdf>
- [9] 'Gateways to Public Digital Learning: Making Digital Education a Public Good'. [Online]. Available: <https://www.unesco.org/sites/default/files/medias/fichiers/2023/12/Gateways%20to%20Public%20Digital%20Learning%20Initiative0.pdf>
- [10] 'UNESCO-UNICEF Charter for Public Digital Learning Platforms'. [Online]. Available: https://www.unesco.org/sites/default/files/medias/fichiers/2025/09/UNESCO-UNICEF%20Charter_Version%20for%20presentation%20to%20Gateways%20countries.pdf
- [11] 'Using Service Workers - Web APIs | MDN', MDN Web Docs. Accessed: Dec. 30, 2025. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/Service_Worker_API/Using_Service_Workers
- [12] J. A. Kulik and J. D. Fletcher, 'Effectiveness of Intelligent Tutoring Systems: A Meta-Analytic Review', *Review of Educational Research*, vol. 86, no. 1, pp. 42–78, Mar. 2016, doi: [10.3102/0034654315581420](https://doi.org/10.3102/0034654315581420)
- [13] 'Guidance for generative AI in education and research | UNESCO'. Accessed: Dec. 30, 2025. [Online]. Available: <https://www.unesco.org/en/articles/guidance-generative-ai-education-and-research>
- [14] 'Ethics of Artificial Intelligence | UNESCO'. Accessed: Dec. 30, 2025. [Online]. Available: <https://www.unesco.org/en/artificial-intelligence/recommendation-ethics>
- [15] 'Emerging governance of generative AI in education: OECD Digital Education Outlook 2023', OECD. Accessed: Dec. 30, 2025. [Online]. Available: <https://surl.li/frkekq>
- [16] S. Grinschgl and A. C. Neubauer, 'Supporting Cognition With Modern Technology: Distributed Cognition Today and in an AI-Enhanced Future', *Front. Artif. Intell.*, vol. 5, July 2022, doi: [10.3389/frai.2022.908261](https://doi.org/10.3389/frai.2022.908261)
- [17] 'Offline data', web.dev. Accessed: Dec. 30, 2025. [Online]. Available: <https://web.dev/learn/pwa/offline-data>
- [18] 'Getting Started: Server and Client Components | Next.js'. Accessed: Dec. 30, 2025. [Online]. Available: <https://nextjs.org/docs/app/getting-started/server-and-client-components>
- [19] 'Guides: PWAs | Next.js'. Accessed: Dec. 30, 2025. [Online]. Available: <https://nextjs.org/docs/app/guides/progressive-web-apps>
- [20] 'Web Storage API - Web APIs | MDN', MDN Web Docs. Accessed: Dec. 30, 2025. [Online]. Available: https://developer.mozilla.org/en-US/docs/Web/API/Web_Storage_API
- [21] 'Streaming'. Accessed: Dec. 30, 2025. [Online]. Available: <https://vercel.com/docs/functions/streaming-functions>
- [22] V. Aleven, I. Roll, B. M. McLaren, and K. R. Koedinger, 'Help Helps, But Only So Much: Research on Help Seeking with Intelligent Tutoring Systems', *Int J Artif Intell Educ*, vol. 26, no. 1, pp. 205–223, Mar. 2016, doi: [10.1007/s40593-015-0089-1](https://doi.org/10.1007/s40593-015-0089-1)
- [23] C. Rutledge et al., 'Gamification in Action: Theoretical and Practical Considerations for Medical Educators', *Academic Medicine*, vol. 93, no. 7, pp. 1014–1020, July 2018, doi: [10.1097/ACM.0000000000002183](https://doi.org/10.1097/ACM.0000000000002183)
- [24] M. Malone, Y. Wang, K. James, M. Anderegg, J. Werner, and F. Monrose, 'To Gamify or Not?: On Leaderboard Effects, Student Engagement and Learning Outcomes in a Cybersecurity Intervention', in *Proceedings of the 52nd ACM Technical Symposium on Computer Science Education, Virtual Event USA: ACM*, Mar. 2021, pp. 1135–1141, doi: [10.1145/3408877.3432544](https://doi.org/10.1145/3408877.3432544)
- [25] J. Krath, L. Schürmann, and H. F. O. von Korfflesch, 'Revealing the theoretical basis of gamification: A systematic review and analysis of theory in research on gamification, serious games and game-based learning', *Computers in Human Behavior*, vol. 125, p. 106963, Dec. 2021, doi: [10.1016/j.chb.2021.106963](https://doi.org/10.1016/j.chb.2021.106963)
- [26] M. Sailer, J. U. Hense, S. K. Mayr, and H. Mandl, 'How gamification motivates: An experimental study of the effects of specific game design elements on psychological need satisfaction', *Computers in Human Behavior*, vol. 69, pp. 371–380, Apr. 2017, doi: [10.1016/j.chb.2016.12.033](https://doi.org/10.1016/j.chb.2016.12.033)